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ARTICLE



A potential screening method for epigenetic drugs: uncovering stress-induced gene silencing in *Chlamydomonas*

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ABSTRACT

Histone modifications and DNA methylation together govern promoter availability, thereby influencing gene expression. This study queries the unicellular chlorophyte, *Chlamydomonas reinhardtii* using a three step “epigenetic assay” design to phenotypically track the variegation of a randomly integrated Paromomycin resistance transgene(s) (Pm^R). Based on its position of integration, the Pm^R gene expression hinged on two epigenetic hallmarks: the spreading of heterochromatin, and the transmissible memory of epigenetic states across generations. Using a spot-dilution analysis, the loss of antibiotic resistance phenotype was scored from 0 to 4, four being maximally silenced. Appropriate construct designs were used to demonstrate that the *cis*-spread of heterochromatin could be interfered with a stronger euchromatic barrier (*TUB2* promoter). When assayed for metal ion stress, a combination of Mn deficiency with excess Cu or Zn stress was shown to induce gene silencing in *Chlamydomonas*. Cu stress resulted in the accumulation of intracellular ROS, while Zn stress elevated the sensitivity to ROS. As proof of functional conservation, mammalian epigenetic drugs demonstrably interfered with stress-induced gene silencing. Finally, a selected group of transgenic clones responsive to HDACi sodium butyrate, when tested in a gradient plate format showed similarity in phenotype to the plant-derived compound cinnamic acid. This indicated a possible commonality in their mode of action, unlike curcumin which might have a different mechanism. Thus, using binned libraries, based on a common set of responses to known drugs, a cost-effective high-throughput screening strategy for epigenetically active compounds from plants or other sources is described.

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Introduction

Epigenetic modulation of genes is dictated by multiple chromatin remodeling machineries that either alter large-scale packaging of the genome or mediate local nucleosomal changes at gene promoters [1]. Altering the nucleosome (basic unit of epigenetic activity) is achieved by three major pathways, namely histone modifications, histone variants, and DNA methylation, each involving a multitude of structural proteins and enzymes that can modulate the way DNA is occluded from the transcription machinery [2,3]. To exemplify some, histone acetylation (e.g. H4K16Ac) and H3K4 trimethylation (me3) correlate with open chromatin (euchromatic); whereas DNA methyltransferases (DNMTs), histone deacetylases (HDACs), and modifications such as H3K9me3/H3K27me3 are more correlated with gene repression or silencing (heterochromatin) in higher eukaryotes [4].

Gene silencing mechanisms and epigenetic screens

Abiotic stress responses in plants (salt, heat, drought, etc.) are mediated by RNAi silencing of specific genes, and *via* DNA methylation or histone modifications in chromatin [5]. Yet unidentified in plants, ROS-mediated epigenetic shutdown silences tumor suppressors in humans, likely selecting for oncogenic mutations thereby predisposing us to diseases like cancer [6,7]. Understanding this broader mechanism of environmentally silenced genes (*via* interaction of HDAC/Polycomb/DNMTs) [6] can be exploited to screen epigenetic drugs which can help us combat either creation or propagation of these faulty chromatin-based memories [3]. Most *in vivo* epigenetic screens target cancer DNA methylation by using locus de-repression assays that report on genomic methylation (RGMs) proximal to promoters of specific candidate genes [8–10]. Besides ongoing structure-based drug design, similar *in vitro*

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library-based screens have led to the discovery of small molecule inhibitors of Polycomb and HDACs [11,12]. These approaches query drugs, either specific to the mode of action of the enzyme or mode of recruitment to a particular promoter/gene. Unbiased functional screens that query new interactions or integrative possibilities among epigenetic pathways seem scarce. An evolutionary understanding of the common denominator of these epigenetic mechanisms across diverse models will not only provide fundamental insights into disease but also an effective toolbox for drug discovery, for these interacting pathways. Although HDACs have been recorded in many organisms, fewer simpler models demonstrate an active involvement of Polycomb (e.g. PRC2/4 in *Drosophila* and zebrafish [13]) or a culmination of all the three pathways including DNA methylation (e.g. CpG aberrations, common in cancer [14]). Although mired in some controversy about functionality, algal cells seem to be one of the earliest primitive organisms to have branched off and yet maintain these three distinct pathways [15].

Chlamydomonas reinhardtii, a haploid unicellular chlorophyte, harbors both histone and cytosine DNA methylation [16]. Statistical analyses of genome-wide DNA methylation patterns in this model have not uncovered correlations of CpG positioning to gene expression, although exons and transposable elements do seem to harbor higher levels of methylation, akin to mammalian counterparts [17]. Contrary to classical site-specific methylation detection assays (MSRE), more recent high-resolution candidate gene studies (using bisulfite sequencing) have implied a repressive role for cytosine DNA methylation in *Chlamydomonas* [18]. Although few, these observations point to an emerging role for DNA methylation and endonucleases in this algal model, probably acting as a defense mechanism to silence foreign DNA (like transposons/viruses, akin to prokaryotes) [19]. In addition to nucleosomal mechanisms [20], the Polycomb subunit *E(z)* (homologue of the catalytic subunit of PRC2/4, *hEZH2*) [15], and RNAi pathway of gene silencing has also been demonstrated in *Chlamydomonas* [21], thereby making it an attractive model to study the functional conservation of epigenetic mechanisms in a haploid unicellular system.

Transgenes and factors affecting gene silencing

Although reporter genes have been widely used for the characterization of tissue-specific gene expression in multicellular models [22,23], haploid unicellular organisms (like yeast and *Chlamydomonas*) offer unique advantages, allowing easy tracking of recessive or co-

dominant traits [18]. Antibiotic resistance transgenes as reporters have been particularly effective in tracking changes in gene expression, simply discernible by growth phenotypes in *Chlamydomonas* [24]. Yet, fewer algal studies have used these to assay the main tenets of epigenetics: (i) the ability of heterochromatic marks to spread to neighboring nucleosomes (classically described in *Drosophila* as position effect variegation, PEV), and (ii) a mechanistic implication of this spreading, i.e. the heritability of these epigenetic memories across generations [25,26]. Several histone modifications of *Chlamydomonas* (including histone methylation and acetylation) have been detected by mass spectrophotometry, but data regarding their genome localization *vis-à-vis* the enzymes involved are scarce [27]. Evaluating the differential contribution of each epigenetic pathway that dictates regulation at specific genomic locations will prove to be critical in designing functional epigenetic screens in *Chlamydomonas* akin to its distant human cousins [9,10].

To enable tracking of silencing pathways in *Chlamydomonas*, we set out to design an unbiased phenotype-based screening assay that would account for both the spread of heterochromatin and the inheritability of epigenetic memories. Since genetic manipulations in *Chlamydomonas* to create knockouts/knockdowns are still quite laborious, such an assay would also allow us to test the functional conservation of pathways by using well-classified inhibitors/drugs (e.g. histone deacetylase – HDACi or DNA methylation inhibitors – DNMTi) commonly used in mammalian systems [28]. Using this design, the study aims to query if oxidative stress-induced epigenetic changes (as observed in cancer) are an evolutionary ancestral yet functionally conserved pathway in the unicellular model *Chlamydomonas*.

Materials and methods

Strains, culture, and growth conditions

Chlamydomonas reinhardtii strains cc124, cc125 (for autolysin preparation), and cc400 used in this study were obtained from the *Chlamydomonas* Resource Center, MN. All cultures were grown in TAP medium under continuous light at 25 °C till it reached a density of 3–4 million cells/mL, and transformed by *Scal* linearized plasmid DNA using glass bead methodology, as previously described [29].

Reagents

All reagents used in the study were bought from Merck or Thermo Fischer Scientific Ltd. (Waltham, MA) unless

mentioned otherwise. All enzymes used in the study were purchased from New England Biolabs (NEB, Beverly, MA). *S*-Adenosyl methionine (SAM) was procured from Sigma (St. Louis, MO). DNA primers were synthesized by Bioserve Biotechnologies, Hyderabad, India. Taq polymerase and PCR reagents were obtained from NEB, Beverly, MA.

The epigenetic assay

Step A: single colonies of the transgenic clones of interest were grown as a patch on 1.5% TAP agar with paromomycin (10 µg/mL, Hi-Media, Mumbai, India). These cells were then inoculated in TAP liquid media (step B) and grown for 3–4 days ($\sim 3\text{--}4 \times 10^6$ cells/mL). At step C, these were serially diluted and spotted on TAP or TAP + paromomycin (designated PARO) agar plates. Since our assay hinges on the inhibition threshold by paromomycin, the antibiotic-sensitive parental strain cc124 is always included on the plates as a control, in order to verify antibiotic activity under the assayed conditions. Plates were incubated under continuous light. The silencing score was quantitated by comparing the number of spots that disappeared on Paro compared to TAP, each spot adding a value of 1. If a spot had countable colonies, and contained less than 50% of its control spot, it would be scored at 0.5. All spotting was carried out in duplicates.

The $1\times$ concentration of Zn and Cu in $1\times$ TAP is 76 µM and 6.3 µM, respectively. Depending on the design of the experiment, the metal ion stress was given either at step B or C. DAC (decitabine, 5-aza-2'-deoxycytidine) was spread on the plates 4 h prior to use, at a final concentration of 1 µM, while sodium butyrate (4 mM) was incorporated into the media. Multiple freeze thaw cycles were avoided in handling DAC as it could get inactivated. For the gradient plate assay, curcumin and quercetin (SRL chemicals) stocks were prepared in ethanol (final working concentration 20 µM and 30 µM, respectively) while sodium butyrate was prepared in water (working concentration 6 mM). Cinnamic acid (1 mM, SRL Chemicals, Chennai, India) was directly dissolved as powder in warm TAP agar, before pouring the plates.

Molecular Methods

DNA extraction and gel electrophoresis

DNA was extracted as described [30]. Both agarose (2%) and polyacrylamide gels (6%) were cast in $0.5\times$ TBE and stained with ethidium bromide for visualization of DNA bands using a BioRad transilluminator, or stained

by silver staining protocol of PAGE gels as described [31].

Creating transgenic clones: Plasmid transformation (in *Escherichia coli* DH5 α), DNA extraction, and *Chlamydomonas* genomic DNA extraction was carried out as described [30,31]. Restriction enzyme digestion and inactivation, Klenow fill-in, DNA ligation, PCR gel purification (Qiagen, Valencia, CA) and *M.SssI* DNA *in vitro* methylation (NEB, Beverly, MA) of the PCR amplicon, were all executed as per manufacturer's specifications. Transformation of *Chlamydomonas* was carried out by glass bead transformation methodology as described in strains cc124 and cc400 [32].

Dot Blot and Western detection: Total protein was isolated from *Chlamydomonas* wild-type and transgenic clones and 10 µg of the total protein was spotted on nitrocellulose membranes and probed with a α GFP antibody, as described by Motiwalla et al. [29].

MSRE assay: 200 ng of template genomic DNA or PCR amplicon was digested with 1 U of enzyme (*HpaII* or *MspI*) as per manufacturer's specifications (NEB), following which the enzymes were heat inactivated. The DNA was then assayed by PCR amplification using primers as detailed in Supplemental Table 1, and annealed at 60 °C, extended at 68 °C for 1 min for a total of 32 cycles using an Eppendorf PCR machine.

DCFDA fluorescence detection: *Chlamydomonas* cells were washed once with sterile water and incubated in 20 µM 2',7'-dichlorofluorescein diacetate (DCFDA, Sigma, St. Louis, MO) for 5 min in the dark. The stained cells were recovered by centrifugation and resuspended in $0\times$ TAP (minus Mn, Cu, Zn) before fluorescence measurement: either by fluorimeter, or using a Nikon fluorescence microscope using a band pass filter [33].

Results

The combinatorial complexity of the "epigenetic code" controls different shades of gene expression ranging from strongly silenced heterochromatin to active euchromatin. In order to design a cost-effective screening strategy for silenced chromatin states in *Chlamydomonas*, we decided to track position effects of a randomly integrated antibiotic resistance transgene as a phenotypic model for gene silencing, as exemplified by earlier studies [15,18]. It was important to use a transgenic form of enzyme-based resistance so that one can minimize other intrinsic factors (e.g. ribosome binding, alternate compartmentalization, etc.) that could compound our inferences.

pCG vector design

The pChlamiRNAi3-derived pCC1G vector contained the divergently transcribed antibiotic selection marker (paromomycin resistance transgene, *Pm^R*) and *CDPK1* gene tagged with a *CrGFP* reporter [29]. To improve on transformation frequencies, we excised the *NdeI-EcoRI* fragment containing the 5'UTR-*CDPK1* region, leaving the divergently transcribed *HSP70:RBCS2::Pm^R* gene and the *PSAD::GFP* to circularize and form a ~7.04 kb pCG construct (Supplementary Figure S1(A)). The loss of *NdeI* and *EcoRI* sites due to Klenow fill-in and blunt-end ligation was confirmed in two separate pCG clones [clones 1 and 3, Supplementary Figure S1(C), *XbaI* digest as positive control]. Clone 3 pCG DNA was linearized with *ScaI* (cuts *amp^r* gene) so as to direct the genomic integration away from the antibiotic resistance reporter *Pm^R* (Supplementary Figure S1(D)). A battery of randomly integrated stable transformants was obtained in both cell wall plus cc124 as well as cw- cc400 strains of *Chlamydomonas*. About 300 transgenic clones were generated and assayed for their epigenetic phenotypes using the design described below.

The epigenetic assay

Multiple stable transformants were derived using the *ScaI*-linearized pCG plasmid, with the possibility of each clone being random, not only in the site of integration but also copy number (either per site or at multiple sites). Since transgenic clones stabilize their genetic background upon multiple passaging (>30 used here), any phenotypic variegation in short-term experiments (<8 days), would most likely be a result of heterochromatin spread, which in turn would be dependent on the neighborhood of integration (position effect). A three-step “epigenetic assay” was designed to detect this cumulative effect of heterochromatin spread on transgene expression, phenotypically. As shown schematically (Figure 1(A)), at step A, cells were grown on paromomycin-containing TAP medium (PARO), thereby selecting the initial background of the *Pm^R* gene to an “ON” state. Next, in step B, these cells were propagated in liquid culture for 4 days (8–10 generations) in the absence of the selection pressure (TAP). Depending on the kinetics of heterochromatin spread, each clone would accumulate a certain percentage of variegated cells where the gene would now be repressed (“OFF” state), making them antibiotic-sensitive. Step B, therefore, represents the *de novo* spread of heterochromatic marks on to the active *Pm^R* gene. By spotting increasing dilutions of these variegated populations at step C on TAP (viability control) versus PARO medium (growth test

on paromomycin, Figure 1(B)), the assay enables us to estimate that fraction of the population, which got silenced as a result of epigenetic shut-down (on a scale of 0–4). The assay, thus, measures the “*de novo*” spread of marks as an initiation of epigenetic memory in step B, which in turn depends on maintenance of the mark and, therefore, the phenotype at step C. By exposing the cells to epigenetic drugs at different steps, we could potentially interfere either with the initiation (drug exposure in step B) or the physiological effect of reversing epigenetic maintenance (if drug treatment is in step C). This ability to test initiation or maintenance of the repressive phenotype, against a constant genetic background, constitutes an important hallmark of epigenetics, that is, heritable states of gene expression based in chromatin mechanisms and not due to mutations. The possibility of spontaneous mutations affecting the specific phenotype will be rare, considering that the assay spans only two 4-day growth cycles (~16–18 population doublings) [34].

Epigenetic phenotypes and effect of drugs

Quantitating the epigenetic variegation using serial dilutions (spot-dilution assay, Figure 1(B,C)) allows us to generate a silencing scale from 0 to 4, where 0 describes an unvariegated clone (possibly depicting integration in a more euchromatic neighborhood, e.g. clones 2, 6, and 10, Figure 1(C)), while a score of 4 depicts the entire population as silenced, posing an active shut down of gene expression or a highly heterochromatic neighborhood (e.g. clones 17 and 111, Figure 3(A)). A majority of the random integrations occurred in open chromatin (~67%) as depicted by their low silencing scores. The percentage of clones that get progressively silenced followed a mathematical decay ($R^2 = 0.9565$, Supplementary Figure S2), indicating that if such a screen approached saturation, it would be possible to compute the total number of epigenetic neighborhoods in the *Chlamydomonas* genome. Our aim was to be able to generate a battery of epigenetically responsive clones that can demonstrate pathway-specific phenotypes and, therefore, serve as a useful screening tool for novel epigenetic compounds.

To obtain the proof-of-principle of epigenetic interference, we tested several clones at step C, either in the presence of a HDACi or DNMTi. Since both sodium butyrate (NaB) and cytosine analogue 5-aza-2'-deoxycytidine (DAC, Decitabine) used here are well documented to affect mammalian cancers [28], the reverse principle of this effect in *Chlamydomonas* offers hints to the conservation of epigenetic mechanisms of gene

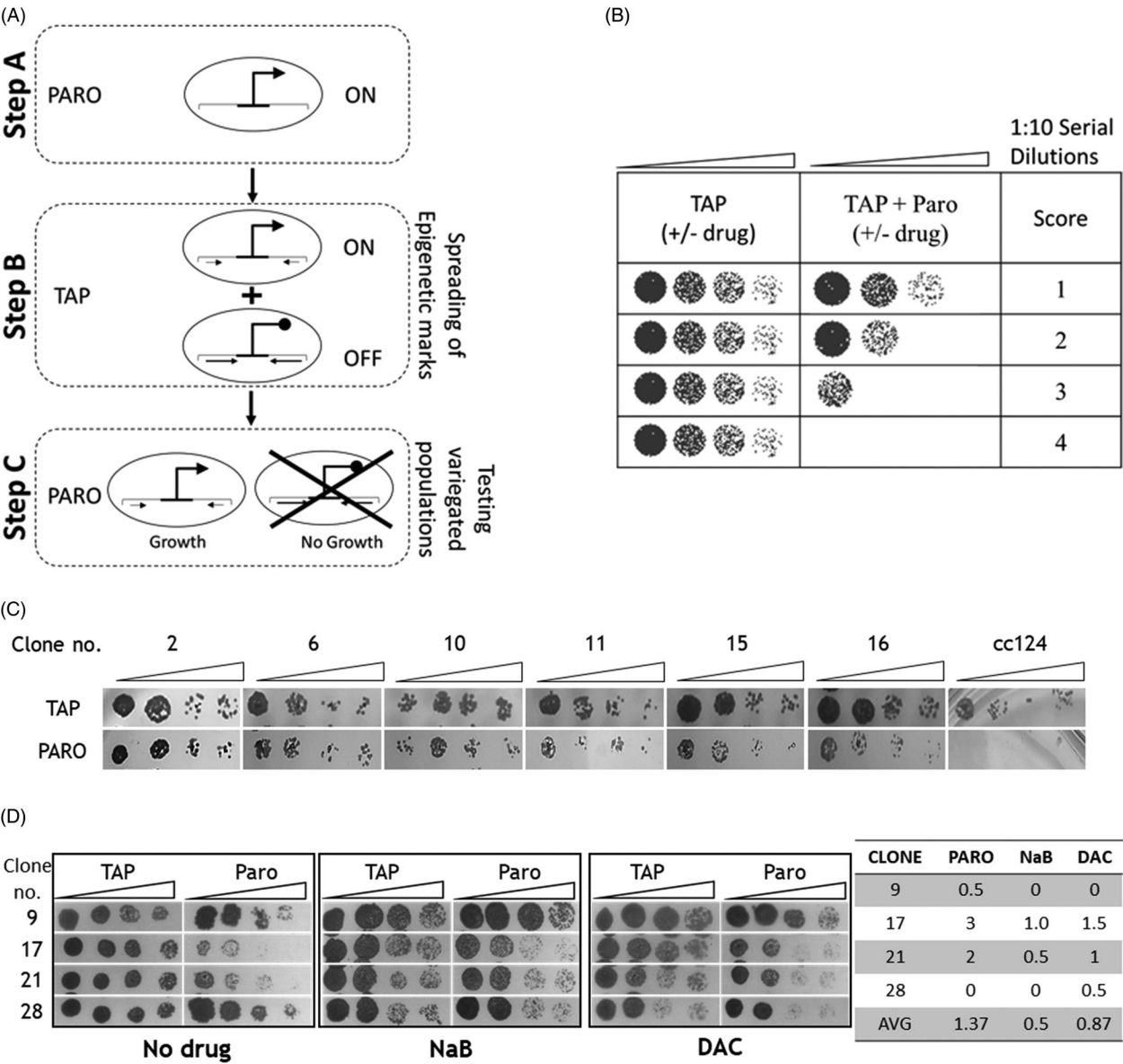


Figure 1. The epigenetic assay. (A) Schematic representation of the three-step assay: At step A, cells are grown on paromomycin antibiotic selection, followed by step B where the selection pressure is removed to allow variegation in the population, and step C where the variegation of gene expression is tested by a spot dilution semi-quantitative growth assay on TAP (control) versus TAP + Paro (test) to account for the population that has been silenced. (B) A growth schematic of the spot dilution assay (from left to right, undiluted, 1:10, 1:100, 1:1000), representing the semi-quantitative “silencing score” derived from the number of spots that disappear in the test (TAP + Paro) compared to the control (TAP, scoring details in methods). (C) About ~300+ pCG clones were screened by the epigenetic assay and plated at step C, to test a comparison of growth viability on TAP versus antibiotic resistance on TAP + paromomycin (Paro). Data of six transgenic pCG clones are represented with the parental cc124 strain as control. (D) Select clones were grown and plated with or without epigenetic drugs (at step C): TAP or Paro containing sodium butyrate (4 mM, NaB) or Decitabine (0.5 μM, DAC). The silencing scores computed from duplicates are represented in the adjoining table along with the averages from the four clones.

silencing, even in this unicellular chlorophyte. Silencing seems to be partially relieved by the two drugs used in pCG clones 17 and 21, reiterating the deduction above (Figure 1(D)). In summary, our assay underlines the following effects: (a) at concentrations that are effective in human cell lines, DAC or NaB is not toxic to *Chlamydomonas*, (b) algal cells seem to be able to

uptake and incorporate the drugs physiologically to affect phenotypes, and (c) inhibition of DNA methylation or histone deacetylation does alleviate transgene silencing phenotypes in *Chlamydomonas*. Although mild, some clones showed increased silencing as a result of these epigenetic inhibitors (e.g. clones 9 and 28, Figure 1(D)). This seemingly opposite outcome was

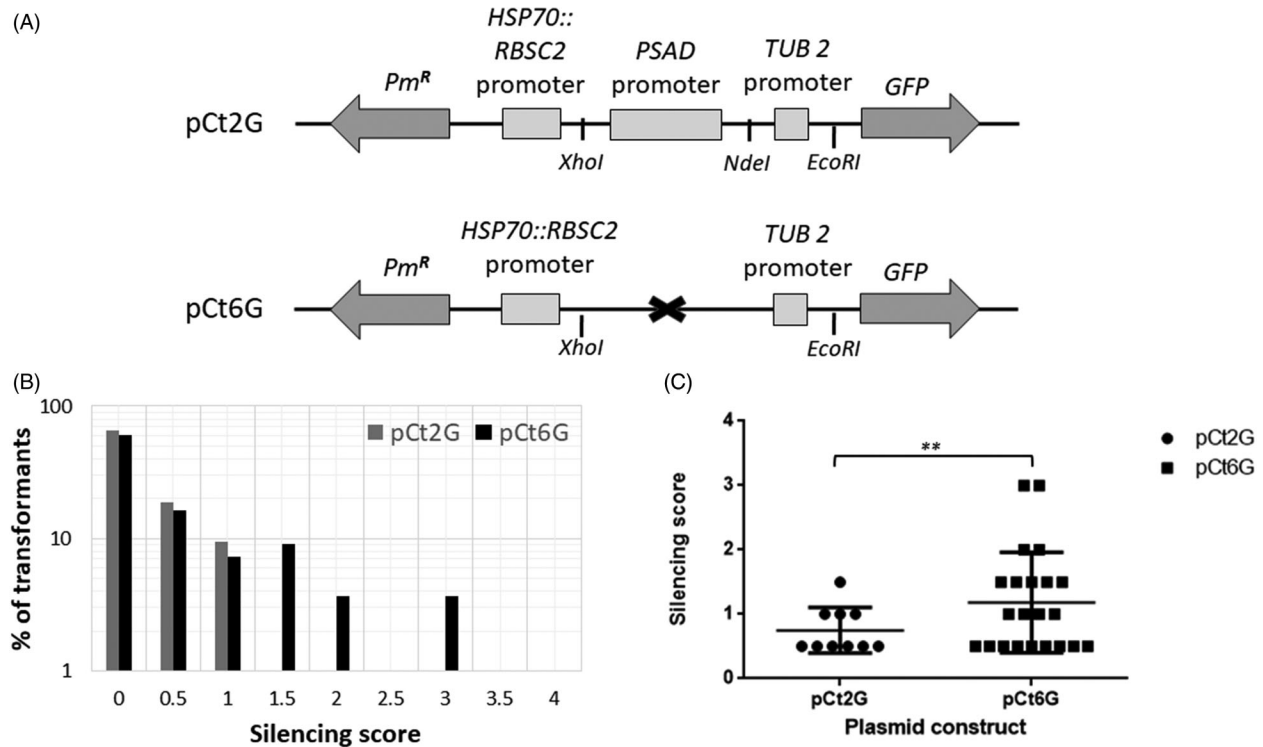


Figure 2. Spread of silencing across a euchromatic barrier. (A) A schematic representation of the vector design (pCt2G and pCt6G) that differs in the PSAD promoter. One must note that both the PSAD and TUB2 promoters are transcriptionally divergent (driving GFP) compared to the HSP70:RBSC2 promoter that drives the *Pm^R* gene. (B) Percentage of the total number of transgenic clones analyzed by the epigenetic assay for constructs pCt2G ($n = 32$), and pCt6G ($n = 55$) plotted against the silencing scale of 0–4. (C) A cluster plot analysis of the epigenetically active clones (0.5–4, silencing score) from the two construct backgrounds indicated a statistically significant difference, as analyzed by an unpaired *T*-test with Welch’s correction (** p -value = 0.039).

probably mediated by epigenetic relieving of proteins/pathways (at secondary locations) that mediate silencing of the *Pm^R* gene, and are, therefore, indirect effects.

Whether these phenotypes translate directly to detectable local molecular changes on nucleosomes is currently under investigation. Methylation sensitive restriction enzyme (MSRE) assay exploits the differential sensitivity of select isoschizomer pairs [*MspI* (^mCH sensitive) and *HpaII* (^mCpG sensitive)] to cleave methylated DNA. Using control DNA that was methylated with a CpG specific methylase, a PCR-based format was standardized to query for CpG methylation changes at the promoter (*SAD1* amplicon, [Supplementary Table S1](#)). As demonstrated, if the amplification is carefully restricted within the sensitive (linear) range, the primers designed across the RE site are able to report on differences in cutting efficiency ([Supplementary Figure S4\(A\)](#)). The same RE pair was tested on genomic DNA of different epigenetic clones [clone no. 5 being least silenced, clones 70 and 28 being intermediate and clone 17 being maximally silenced (inset, [Supplementary Figure S4\(B\)](#))]. The digested DNA was tested by PCR of the transgenic *HSP70:RBSC2* promoter ([Table S1](#) and [Figure S1\(A\)](#)). Compared to the undigested control, *MspI* restriction (CHG context) was unaffected and displayed more or less

equal efficacy across all four clones ([Supplementary Figure S4\(B\)](#)). *HpaII* restriction (CpG context) on the other hand was relatively protected in clones that showed higher silencing ([Supplementary Figure S4\(B\)](#)). Withstanding the limitations of both MSRE and end point PCR, our results indicated that DNA methylation probably has some role in transgene repression at driving the *HSP70:RBSC2* promoter. Since the rare occurrence of restriction sites limits query only to a few cytosine residues, such methylation differences need to be verified with high-resolution bisulfite sequencing technologies.

In parallel, several attempts to visibly track the divergently transcribed *CrGFP* under *PSAD* control remained unsuccessful (by fluorescence microscopy, spectroscopy, western blot; unpublished results), probably due to the deletion of the 5’UTR which reduces translation efficiency (dot blot, [Supplementary Figure S3](#)).

Euchromatic barriers can affect the spread of epigenetic states

Despite the *Pm^R* gene being driven by one of the most robust artificial fusion promoters described in *Chlamydomonas*, namely *HSP70:RBSC2*, we were still

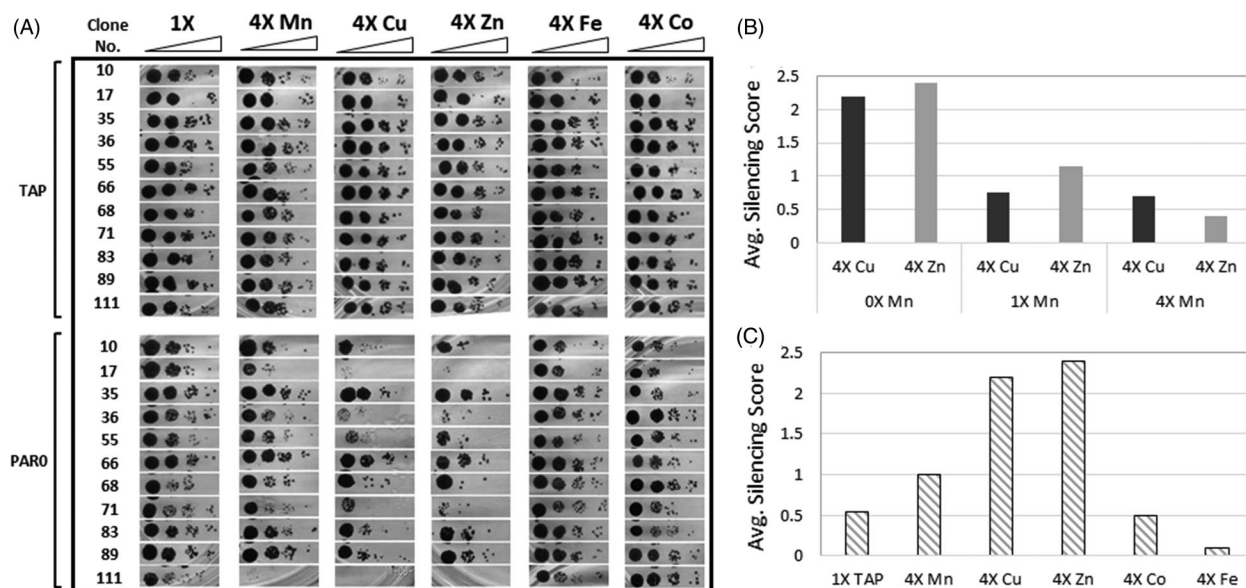


Figure 3. Metal ion stress-induced silencing. At step C of the *Epigenetic Assay*, pCG clones were plated on either Mn^+ or Mn^- media containing four times excess concentration of respective metal ions (4 $\times$) compared to the control (1 $\times$). The 1 $\times$ concentration of ions were Mn – 31 μ M, Cu – 6.3 μ M, Zn – 76 μ M, Co – 6.8 μ M, and Fe – 18 μ M. Panel A depicts the spot dilution assay results (step C) on Mn^- media: TAP (control) versus Paro (test), as detailed in the Methods. The cumulative responses of the eleven clones tested were plotted as average silencing scores in panel B (a comparison of Cu and Zn stress as a function of Mn) and panel C (a comparison of all five metal ions tested in panel A).

able to select for epigenetically active clones that show variegated antibiotic resistance phenotypes. This indicated that there must be strong heterochromatic neighborhoods in the genome that could silence even such active promoters. Since epigenetic mechanisms invoke the *cis*-spread of epigenetic states, we tested if a barrier could block or slow down the spread of heterochromatin onto the Pm^R gene. We created two new constructs which differed in the presence or absence of a strong *PSAD* promoter along with the previously reported tubulin promoter (*TUB2*, 5' to *CrGFP*, Figure 3(A)) [35]. A 393 bp region of the *TUB2* promoter was PCR amplified (primers designed with appropriate restriction sites, Table S1) and ligated either into the *NdeI-EcoRI* site or the *XhoI-EcoRI* site of the excised pCC1G backbone vector (Supplementary Figure S1(A)) to form two new plasmids: pCt2G (*PSAD* + *TUB2*) and pCt6G (*TUB2* alone, Figure 2(A)), respectively. Epigenetic analysis showed that pCt6G transformants, lacking the *PSAD* promoter, underwent silencing more often compared to pCt2G (containing *PSAD*, Figure 3(B)). This indicated that *PSAD* + *TUB2* possibly nucleates a stronger active state, which might act as a euchromatic barrier to the spread of heterochromatin toward the *HSP70:RBCS2* promoter that drives the visible phenotype (thereby leading to lower silencing scores in pCt2G derived clones). One must note that the direction of transcription by the *PSAD* + *TUB2* promoter is divergent from Pm^R . Therefore, although unlikely, any increase of Pm^R gene

transcription, if mediated by *PSAD* + *TUB2* directly, should have been uniform across different transgenic clones (intrinsic factor). However, since we observed different phenotypes even among pCt2G-derived clones themselves (Figure 2(A)), the variegation could be attributed to an extrinsic factor; in this case, the possible *cis*-spread of heterochromatin. In summary, this genetic design adds credence to the conservation of epigenetic mechanisms of spreading of repressive marks *via cis*-acting chromatin elements (position effects).

Stress-induced gene silencing

Low levels of intracellular reactive oxygen species (ROS) are well documented to exert physiological control on gene expression, both *via* light-dependent and -independent pathways [36]. However, a more direct interaction of ROS-induced gene silencing *via* epigenetic proteins was first described in mammalian models of cancer [6,37]. We wanted to query if this is functionality conserved, even in *Chlamydomonas*. Besides harboring several ROS handling systems, *Chlamydomonas* requires a number of trace metal ions for growth, namely transition metals like Cu, Mn, Co, and Fe and the post-transition metal Zn [38]. Some of these metal ions are known to generate ROS through Fenton chemistry [33]. Can metal ion stress and the resulting ROS prove relevant in creating epigenetic memories *in vivo*? We assayed both

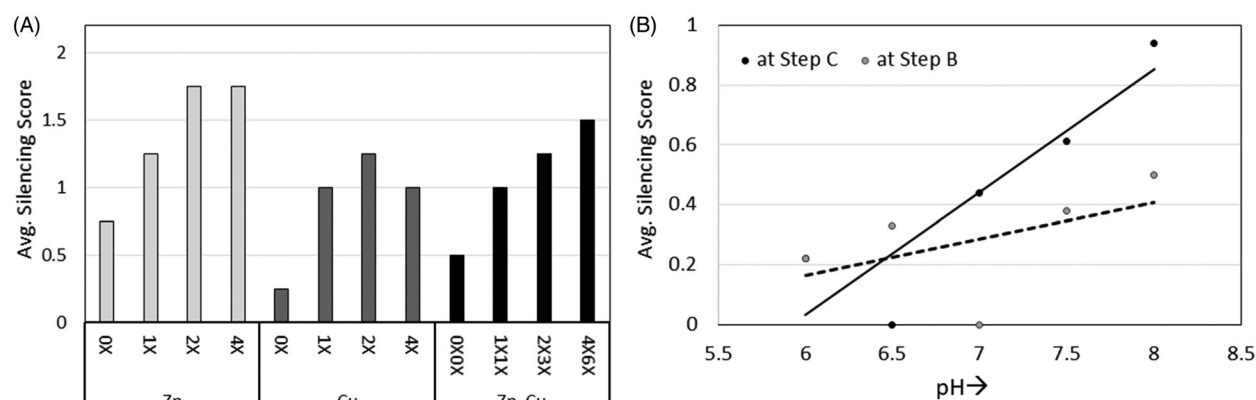


Figure 4. Memory of abiotic stress. To test for the memory of ion-exposure on the epigenetic phenotype, metal-stress was varied at step B (during growth) followed by the spot-dilution assay (plating at step C) on normal 1X TAP media. (A) At step B of the epigenetic assay (refer Figure 1), pCG clones 17 and 21 were grown (in Mn^- TAP) at different metal-stress concentrations of either excess Zn or Cu or a combination of both (Methods, e.g. 4 $\times$ denotes four-fold excess). After washing the cells free of the excess ions, they were serially diluted and spotted on 1 $\times$ TAP and 1 $\times$ Paro agar plates and scored (at step C). (B) pH dependence: pCG clones from step A (5, 9, 14, 15, 17, and 18) were either grown in 1 $\times$ TAP at varying pH (at step B) followed by plating and scoring on normal 1 $\times$ TAP and 1 $\times$ Paro (gray circles, dashed trend-line); or grown on 1 $\times$ TAP followed by plating at different pH on 1 $\times$ TAP and 1 $\times$ TAP + Paro (at step C, black circles, bold trend-line). The effect of pH on averaged silencing scores of the six clones (for each pH) is plotted as a linear regression.

metal deficient (dropout) as well as excess metal ion stress (4 $\times$ conc., at step C), to query if they affect the epigenetic phenotypes of eleven independent epigenetic clones. Interestingly, although Fe and Co did not show much difference, Cu and Zn stress stood out as having very prominent effects in inducing paromomycin sensitivity (Figure 3(A,B)). We also noted that these effects were particularly light dependent and exacerbated, when cells grew on Mn-deficient media (Mn^- , Figure 3(C)). Based on previous studies of light dependency [39] and the types of metal ions involved indicated that the $Mn^- + Zn/Cu$ stress might mediate either through Photosystem II (which uses Mn at the reaction center) and/or through modulation of Mn-SOD enzymes that are used by *Chlamydomonas* to manage ROS. Repressive effects of Zn could also be attributed to its co-factor function in silencing proteins like DNMTs, Polycomb, and HDACs. As captured by stress-induced repression of a transgene, we suggest that besides down-regulation of specific pathways [39], *Chlamydomonas* exerts broader epigenetic repression in response to Cu or Zn excess.

Memory of abiotic stress

If the above paradigms of abiotic stress are truly epigenetic, they should be heritable as well. In other words, what effect would pre-exposure of the cells to abiotic stress have on the epigenetic output later, when the stress no longer exists? To query this, we tested the epigenetic assay by separating the stress insult (at step B) from the manifestation of the phenotype (at step C).

Since abiotic stress can also lead to cell mortality, care was taken to choose a range of stress conditions that affected cell growth minimally during the assay (as exemplified by their growth curves, Figure 5(B)) [19]. In addition, since the assay had an internal calibration of a viability control on TAP media, overall lethality could be negated out, making the assay specific to monitoring antibiotic sensitivity. Cells grown under increasing concentrations of Zn and/or Cu stress at step B were later tested for epigenetic effects on 1 $\times$ TAP versus Paro at step C (normal growth conditions). We observed that the silencing effects of both Zn and Cu stress could manifest later, even when the abiotic stress had been removed (Figure 4(A)). When tested in combination, we did not observe any significant synergistic effects. pH is known to affect the assimilation of metal ions and enzyme activity. Higher pH seemed to be better at epigenetic silencing when coincident with the phenotype (plating on 1 $\times$ TAP, step C, Figure 4(B)) rather than as a memory (at step B). This acts as a good control for the metal ion assimilation paradigm, as pH is known to be buffered in biological systems and, hence, its effects are likely to be more immediate and not epigenetic in nature, as suggested by the data.

Generation of ROS as a function of metal ion stress

Since previous reports have demonstrated ROS-induced gene silencing in mammalian systems, it was important to query how evolutionarily ancient this pathway might be? This may help not only in the fundamental query of

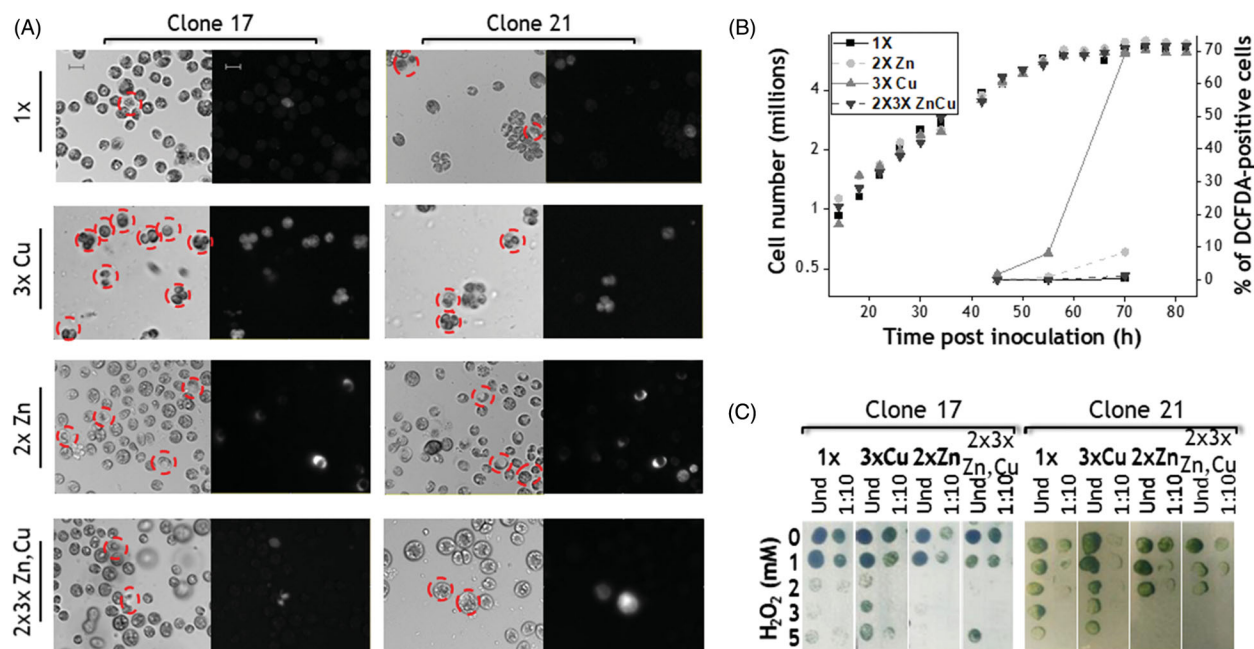


Figure 5. Evaluation of ROS in copper and zinc stress. Clones 17 and 21 were grown either in 1× liquid Mn⁺ TAP media or one containing excess zinc (2× Zn) or copper (3× Cu) or a combination of both (2× 3× Zn,Cu). These cells were assayed by DCFDA staining (see Methods) for intracellular accumulation of ROS. Panel (A) shows bright field (left) and corresponding fluorescence images (right) of DCFDA positivity at 20× (scale bar at top left corner for clone 17-1× = 10 μM). Panel (B) denotes the growth curves for each condition (clones 17) plotted as a function of cell number over time (left axis) overlaid with the quantitation of DCFDA positive cells (right axis), from 45 h (log phase) to 70 h (stationary phase) joined with lines (key as inset) [18]. In addition these cells were also tested for sensitivity to externally added hydrogen peroxide (concentration range 0–5 mM) and assayed for viability differences by spotting dilutions of the peroxide treated cells on 1× TAP (panel C).

epigenetic mechanisms but also design screening strategies for relevant mammalian drugs. In taking clues from Zn/Cu stress and ROS biology [33], we wanted to decipher if these metal ions led to ROS accumulation in our epigenetic paradigm. We used the intracellular ROS indicator DCFDA to stain live cells that were grown under excess Zn (2×) or Cu (3×) or a combination of both (2× 3×). Interestingly, Cu stress led to a high accumulation of DCFDA positive cells as a function of late log phase (Figure 5(A,B)). Although Zn stress barely showed DCFDA positivity compared to control, the combination of Zn and Cu also did not show enough ROS producing cells. This led to a puzzling conundrum, as epigenetic phenotypes were still quite severe under these conditions (Figure 4(A)). Since the DCFDA assay detects ROS only in intact live cells that have esterase activity, there was a possibility that Zn-stressed cells which accumulated high ROS (due to Cu stress) were dying, and, hence, seemed DCFDA negative. When tested for sensitivity to ROS (externally added peroxide), the viability of cells grown in both 2× Zn and 2× 3× ZnCu combination seemed to be more compromised (Figure 5(C), death at 2 mM H₂O₂ compared to 5 mM for control). This Zn-dependent sensitivity to ROS explained

the depletion of DCFDA-positive cells in 2× 3× ZnCu compared to Zn-alone stress (Figure 5(B)). These results put together (Figures 3 and 5) indicated that Cu stress-induced ROS production and eventually led to gene silencing, while Zn stress exacerbated these effects in the 2× 3× ZnCu stress condition, by making those cells more sensitive to ROS damage.

Library design for screening of epigenetic drugs

Although chromatin marks and their enzymes have been identified in *Chlamydomonas* [15], not many drugs have been assayed for epigenetic efficacy. Depending on epigenetic control at the neighborhood of integration, epigenetic variegation of each pCG-transgenic clone might be dominated either by a specific pathway, or a combination thereof (e.g. DNA methylation and/or HDACs). We binned a set of transgenic clones based on their responses to different classes of known inhibitors (with established mechanisms for specific pathways). This binning would help build differential libraries of phenotypically responsive clones, which could help narrow down new compounds toward each mechanism (or pathway). In order to exemplify this, phenotypes

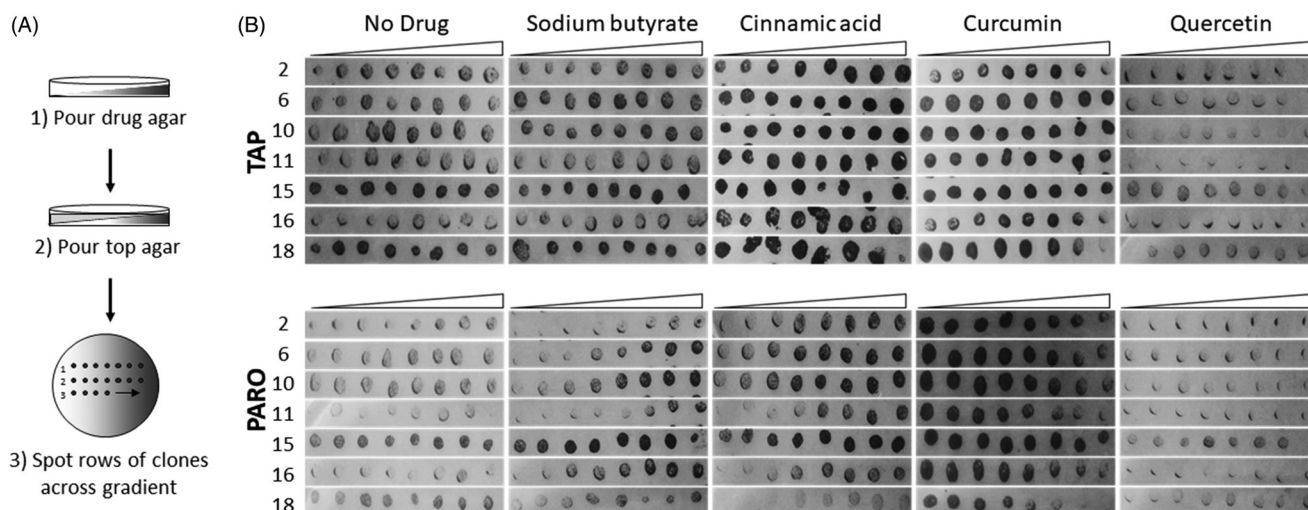


Figure 6. Gradient Plate Assay as an epigenetic screen. (A) A schematic representation of the gradient plate assay design depicting how it was performed. (B) Analysis of multiple pCG clones by the gradient plate assay with the solvent used as the “No Drug” control (only ethanol as solvent gradient control is shown). The top panels are the cells grown on TAP as viability control, compared to the bottom panel (PARO) which tests for antibiotic sensitivity. Since the spots are laid along the drug gradient (see Methods), disappearance of corresponding spots in the direction of gradient in the test is scored as epigenetic silencing.

driven by sodium butyrate (6 mM), an HDAC inhibitor, were compared to three plant-derived compounds (1 mM cinnamic acid, 20 μ M curcumin, and 30 μ M quercetin). Since these epigenetic drugs were previously uncharacterized in *Chlamydomonas*, a range of concentrations derived from those used in mammalian cell systems were tested. A gradient plate methodology was used to test for changes in epigenetic activity at step C; multiple clones were spotted at the same concentration of cells, horizontally as replicates, on a drug gradient (both with and without paromomycin, Figure 6). This would help us discern patterns of silencing observed along the direction of the gradient (Figure 6(A)). Unfortunately, due to its heat-labile nature, we were unable to incorporate the DNA methylation inhibitor (DAC) into the molten agar, for this assay.

When compared to the appropriate control (TAP alone), clones on the “no drug” PARO plates showed different levels of silencing, but no changes along the gradient (Figure 6(B)). The HDAC inhibitor (sodium butyrate) displayed a duality of concentration-based effects for multiple clones (2, 6, 10, 11, 16 and 18, Figure 6(B)) with increased silencing at lower drug concentrations and seemingly relieving silencing at higher concentrations. This effect could be attributed to the low levels of HDACi affecting only a subset of genes, probably enhancing repressive proteins that mediate epigenetic shut down; while higher levels of deacetylation relieving global gene expression as reported by the transgene. In stark contrast to curcumin and quercetin, cinnamic acid showed phenotypic similarity to

butyrate, for a large sub-set of overlapping clones (six out of seven, Figure 6(B)). This indicated that cinnamic acid might potentially act as a deacetylase inhibitor, as was recently described [40]. Quercetin did not show any gradient-specific effects, while curcumin increased silencing only at higher drug concentrations, albeit for a different set of clones (Figure 6(B)). In order to test whether quercetin lacked epigenetic activity in *Chlamydomonas*, or was simply unable to enter the cells due to the cell wall barrier, we tested the same paradigm on cell wall minus mutants (cc400-derived pCG clones). Since very few clones showed drug-dependent epigenetic modulations, our results indicated that quercetin seems to display minimal modulation in *Chlamydomonas* (Figure S5). Curcumin, on the other hand, made these cells sensitive, indicating increased permeability or bio-availability due to the absence of cell wall (Figure S5). In summary, the drug-responsive phenotypes discerned on the gradient plate assay could be evaluated for epigenetic activity, and also provide hints toward the mode of action of the unknown compound, in one single step.

Discussion

Chlamydomonas reinhardtii has functional conservation of multiple silencing pathways, namely histone deacetylases, Polycomb, and DNA methylation [15,41,42]. Since mutants in epigenetic pathways have been rarely characterized in this alga, very little is known about the molecular nature of the players involved [27]. In order

to work around these caveats, we present the design of an “Epigenetic assay,” a phenotypic strategy that would not only query functionality of epigenetic pathways (Figures 1–4), but could also be used to discover new drugs in a phenotypic manner (Figure 6).

Relevance of the epigenetic assay

Mechanisms of epigenetic drugs, like HDACi (e.g. butyrate, valproic acid, etc.) and DNMTi (cytosine analogues, Decitabine, Zebularine, etc.), have been characterized in sufficient detail in mammalian models [43]. However, many of these drugs have not been tested for their efficacy in *Chlamydomonas*. This study uses the “Epigenetic assay” to validate epigenetic hallmarks both phenotypically (Figures 2 and 4) and molecularly (Supplementary Figure S4(B)), as well as with established epigenetic inhibitors (Figures 1(D) and 6). The *cis*-spread of heterochromatin (involved in mechanisms of inheritance (Figure 2) and the modulation of “memories” across generations (Figure 5) opens up avenues for molecular enquiry of epigenetic pathways that function in *Chlamydomonas*. The salient features of the assay are as follows: (a) keeping the genetic background constant in each clonal context (making copy number less relevant), changes in reporter gene expression (depending on the site of integration) can be phenotypically quantitated (Figures 1(C,D) and S2). (b) The variegation could arise either from a single mechanism or an interactive output of multiple epigenetic pathways. Thus, depending on the site of integration of the transgene, specific pathways can be addressed by characterizing phenotypes that respond to specific drugs (Figure 1(D)). As the screen approaches saturation, the clones can be binned into specific “drug-responsive” groups, thereby creating libraries (Figure 6). (c) The assay can test both *de novo* formation of epigenetic memories (at step B, Figure 4), or reverse pre-formed epigenetic memories later (at step C, Figures 1(D) and 6). Unlike most targeted screens that employ homologous knock-in-reporters at specific genetic loci [9,10], this assay presents an unbiased screening strategy. It allows flexibility, not only to address how epigenetic drugs affect heritability but also potentially analyze the epistasis of interactions of different mechanisms (e.g. drug X at step B followed by drug Y at step C). Thus, besides harboring potential as a discovery screen toward predicting their mode of action in a single step, studying interference of memories of one drug with another, holds value in designing therapeutic protocols [44].

Conservation of gene silencing from *Chlamydomonas* to humans

Most of the random integrations of the introduced plasmid (~67%) were biased toward relatively open euchromatic neighborhoods (Figures 2(B) and S2). In this respect, *Chlamydomonas* seems to be more similar to mammalian models, where random insertion by NHEJ recombination is directed toward open chromatin due to better accessibility by the recombination machinery (unlike yeast, where homologous recombination dominates [45–47]). The rest of the clones (~30%) seem to show an almost exponential decrease in frequency of stronger epigenetic phenotypes, indicating the possibility of a finite number of quantifiable heterochromatic neighborhoods in *Chlamydomonas* (Supplementary Figure S2). The role of DNA methylation by MSRE detection (Supplementary Figure S4) and by specific DNMTi (used against human cancers, DAC, Figure 1(D)) implicated a conservation of epigenetic mechanism in *Chlamydomonas* [18,28]. Also captured by HDACi sodium butyrate, this conservation suggests that *Chlamydomonas* can serve as a cost-effective screening model to evaluate drugs that might harbor human potential. In addition, the discovery of ROS induced gene silencing mechanisms (uncovered by Cu or Zn stress, Figures 4 and 5), underlines another evolutionary conservation as originally observed in mammalian models [6]. Cu stress is well established to generate ROS through Fenton chemistry, which in turn causes oxidative protein/DNA damage, leading to cell lethality [33,48]. While both Cu and Zn serve as essential co-factors, besides regulating Cu uptake in *Chlamydomonas* [49], Zn is also important for the repressive functionality of epigenetic proteins in multiple models (e.g. zinc finger domains in DNMTs and CXC-Zn binding motifs in Polycomb [15,50]). This might play an adaptive role to environmental stress [51]. ROS-induced changes in gene expression (in *Chlamydomonas* and higher plants) [36] have mostly focused on the induction of ROS-handling enzymes (GPx and SODs) or shutting down of specific biochemical/photosynthetic pathways, albeit lacking in epigenetic clarity [52]. Transgenic by design, the assay design presented here, by-passes these intrinsic regulatory switches and hinges on position effects (Figure 2). By querying the conservation of ROS-induced gene silencing in *Chlamydomonas* (which sits at the cusp of multicellularity), we might be able to compute the minimal core interactions of epigenetic proteins, crucial toward deriving multicellular complexity or disease.

A one step screening library – detection to function

In the genetically tractable fungus *Neurospora* (unlike yeast and *Drosophila*), DNA methylation is tightly linked with histone modifications, showcasing a conservation of the metazoan epigenetic pathways [53]. In addition to sharing programs of multicellularity common to both plants and animals, similar tractability is offered by the “green yeast” *Chlamydomonas*; wherein epigenetic drug responses, at concentrations similar that of mammalian models, reiterate this conservation (Figures 1 and 6) [54]. Based on the dominant epigenetic pathway affecting position effects, each transgenic clone responded differently (Figure 6). By grouping clones with similar responses against each drug (or pathway), these functionally binned libraries can then be reverse engineered to discover novel drugs in those pathways, maintaining high-throughput potential. Sodium butyrate (HDACi) showed phenotypic similarities to Cinnamic acid (an unsaturated carboxylic acid found in many plants, Figure 6(B)), recently identified not only as an HDACi in humans, but also harboring anti-tumorigenic activity [40]. In this regard, curcumin which shows a pattern different from sodium butyrate (Figure 6(B)), besides acting as an anti-oxidant has also been reported as a DNMTi [55] (not assessed here, due to the lability of the DNMTi drug DAC), and, hence, shows very different epigenetic phenotypes compared to cinnamic acid (for the same library of clones, Figure 6(B)). However, being very pleiotropic in nature, its exact epigenetic function (enhancing silencing at higher concentrations, Figures 6 and S5) in this algal model is yet to be evaluated.

Conclusion

Although epigenetic pathways have been described in multiple metazoans, simpler models like yeast and *Drosophila* do not use DNA methylation. Due to a poor understanding of the protein players involved, epigenetic query in *Chlamydomonas* has been difficult. In search of mechanistic conservation, we propose transgene silencing in *Chlamydomonas* as an unbiased screening model that can be interfered with established epigenetic drugs. Besides acting as an unbiased discovery screen for epigenetic drugs, our study provides the first report towards ROS-induced gene silencing in *Chlamydomonas*, as the common evolutionary denominator that also manifests in cancers.

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Author contributions

SS conceived and executed the experiments with SK, SP, and USJD provided the plasmid pCC1G and strains used in the study. SK contributed to Figures 1–4, SP contributed to Figure 5 and US contributed to Figure 6. SS wrote the paper with inputs from all the authors.

Author agreement

The authors are in informed agreement about the authenticity of the data presented, authorship, and necessary permissions for peer review submission.

Disclosure statement

The authors have no competing personal or financial conflicts of interest to declare that would influence this study.

Statement of informed consent

No conflicts, informed consent, human or animal rights applicable.

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